

EXECUTIVE BRIEF

Title:

Retail-Pulse: E-Commerce Customer Intelligence

Problem Statement:

Instacart operates one of the largest grocery delivery platforms, generating millions of customer transactions. Despite this scale, significant business value is lost through customer churn, declining reorder behaviour, and low-engagement customer segments. This analysis examines over 32 million transactions to identify patterns of revenue leakage, detect customers at risk of churn, and uncover actionable strategies to improve retention and customer lifetime value.

Key Findings:

- Days since last order is the single strongest churn predictor
- The majority of customers belonged to low-engagement segments, while only a small proportion were classified as Champions.
- Produce and dairy have the highest non-reorder counts (3.3M and 1.7M)
- Cohort retention drops sharply after early orders
- Top reordered products are all fresh produce — bananas, organic strawberries
- Monday 10 am is peak ordering time

Recommendations:

1. Re-engagement Campaign for Inactive Customers. Target customers who haven't ordered in 15+ days, since the days since last order is the strongest churn predictor. A push notification or email on day 15 could recover a significant portion of at-risk customers.

2. Loyalty Program for Produce and Dairy. These departments have the highest non-reorder counts. A "subscribe and save" or loyalty discount on fresh produce could convert one-time buyers into repeat customers.

3. Champion Conversion Strategy. Only 8.9% of customers are Champions, but they order an average of 55 times. Focus retention efforts on the 39.3% Loyal Customers segment; they're one step away from becoming Champions with the right incentives.

Tools:

SQL| Python| Power BI